Lecture 39

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1 Limits and Sequences

1.1 Examples of Limits

1. **Squeeze Theorem Application**:

$$\lim_{n\to\infty} \frac{\sin n}{n} = 0$$

Proof: Since $-1 \leq \sin n \leq 1$, we have:

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$$

As $n \to \infty$, both bounds go to 0, so the limit is 0.

2. **Trigonometric Sequence**:

$$\lim_{n\to\infty} n \cdot \pi \cdot \sin(n\pi) = 0$$

Reasoning: Since n is an integer, $\sin(n\pi) = 0$ for all n. The sequence is constantly 0.

1.2 Quiz / Problems

1. Calculate the limit:

$$\lim_{n\to\infty} \frac{2n + n^2}{2025! + n} = +\infty$$

Reasoning: The numerator grows as n^2. The denominator is linear in n (2025! is a large constant). The term n^2 dominates.

2. Standard Factorial Limit:

$$\lim_{n\to\infty} \frac{n!}{n^n} = 0$$

Proof sketch:

$$0 < \frac{1 \cdot 2 \cdot \dots \cdot n}{n \cdot n \cdot \dots \cdot n} = \frac{1}{n} \cdot \frac{2}{n} \cdot \dots \cdot \frac{n}{n} \leq \frac{1}{n}$$

By the Squeeze Theorem, the limit is 0.

2 Construction of Real Numbers and Exponentials

2.1 Real Numbers as Limits

Real numbers can be defined as limits of sequences of rational numbers.

- **Example:** $\sqrt{2} \approx 1.414235\ldots$ Consider the sequence $a_n \in \mathbb{Q}$:

$$1, 1.4, 1.41, 1.414, 1.4142, \ldots$$

This sequence is increasing and bounded above. Therefore, it must converge.

$$\sqrt{2} = \lim_{n\to\infty} a_n$$

2.2 Exponential Functions

We want to define a^x for $a > 1$ and $x \in \mathbb{R}$. We already know a^x for $x \in \mathbb{Q}$ (rational exponents).

2.2.1 Proposition: Monotonicity on $\mathbb{Q}$

The function $f : \mathbb{Q} \to \mathbb{R}$ defined by $f(q) = a^q$ is increasing. **Proof Sketch:** Let $\frac{m_1}{n_1} < \frac{m_2}{n_2}$. This implies $m_1 n_2 < m_2 n_1$. Since $a > 1$, raising to positive integers preserves order. Eventually, this leads to showing $a^{q_1} < a^{q_2}$.

2.2.2 Definition for Real Exponents

To define a^x for $x \in \mathbb{R}$, we approximate x with a rational sequence (q_n) such that $\lim q_n = x$.

- **Example:** $a^{\sqrt{2}}$. Using the sequence approximating $\sqrt{2}$:

$$a^{\sqrt{2}} = \lim_{n\to\infty} \{a^1, a^{1.4}, a^{1.41}, a^{1.414}, \dots\}$$

  Alternatively, using the supremum definition:

$$a^x = \sup\{a^q : q \leq x \wedge q \in \mathbb{Q}\}$$

2.3 Properties of the Exponential Function

Let $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = a^x$.

1. **Algebraic Properties:**

$$a^{x+y} = a^x \cdot a^y$$

$$(a^x)^y = a^{xy}$$

2. **Range:**

$$f(\mathbb{R}) = (0, +\infty)$$

   Proof: For any $y \in (0, +\infty)$, there exists $x \in \mathbb{R}$ such that $a^x = y$ (existence of logarithms).

3. **Case $0 < a < 1$:** Defined via reciprocal:

$$a^x = \frac{1}{(1/a)^x}$$

   Since $1/a > 1$, we use the previous definition.

3 Recursive Sequences and Convergence

3.1 Problem 1: Nested Radical

Consider the sequence defined by:

$$x_1 = \sqrt{2}, \quad x_{n+1} = \sqrt{2 + x_n}$$

Find its limit.

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$$

Solution:

1. **Boundedness:** We show $x_n \leq 2$ by induction.
   - Base case: $\sqrt{2} < 2$.
   - Step: If $x_n \leq 2$, then $x_{n+1} = \sqrt{2 + x_n} \leq \sqrt{2 + 2} = 2$.
2. **Monotonicity:** (x_n) is increasing.

   $$x_{n+1} \geq x_n \iff \sqrt{2 + x_n} \geq x_n \iff 2 + x_n \geq x_n^2 \iff x_n^2 - x_n - 2 \leq 0$$

   The roots of $x^2 - x - 2$ are 2 and -1. Since $0 < x_n < 2$, the inequality holds.
3. **Limit Calculation:** Since the sequence is bounded and increasing, it converges to some limit L.

   $$L = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \sqrt{2 + x_n} = \sqrt{2 + L}$$

   $$L^2 = 2 + L \implies L^2 - L - 2 = 0$$

   Solutions: $L_1 = 2$, $L_2 = -1$. Since limits of positive sequences must be non-negative, $L = 2$.

3.2 Inequalities in Limits

Question: If $a_n > b_n$ for all n, and the sequences converge to A and B, is it true that $A > B$?

Answer: No. The strict inequality is not necessarily preserved in the limit.

- **Counter-example:** Let $a_n = \frac{1}{n}$ and $b_n = 0$. Clearly $a_n > b_n$ for all n. However, $\lim a_n = 0$ and $\lim b_n = 0$, so $A = B$.

Theorem: If $a_n \geq b_n$ (or $a_n > b_n$) and limits exist, then $\lim a_n \geq \lim b_n$.

3.3 Problem 2 (Homework/Exercise)

Consider the sequence:

$$x_1 = 3, \quad x_{n+1} = \frac{1}{4 - x_n}$$

3.4 Heron's Method (Newton's Method for Square Roots)

To approximate $\sqrt{2}$:

$$x_1 = 2, \quad x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$$

This sequence converges to $\sqrt{2}$ very rapidly.